
HOLLY P. O'ROURKE, PhD

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CURRENT ACADEMIC APPOINTMENTS

2025–present: Associate Professor, Department of Psychology, University of California, Riverside

2025–present: Cooperating Faculty Member, Department of Statistics, University of California, Riverside

EDUCATION

PhD, 2016: Quantitative Psychology, Arizona State University

Dissertation: *Time Metric in Latent Difference Score Models*

MA, 2013: Quantitative Psychology, Arizona State University

Thesis: *Mediation as a novel method for increasing statistical power*

BS, 2009: Psychology, Arizona State University

RESEARCH INTERESTS

Quantitative methodology for behavioral science; longitudinal modeling; mediation analysis; latent change score models; zero-inflated count outcomes; machine learning for behavioral prediction; idiographic and nomothetic modeling; applications to health, prevention, education, and developmental science

SELECTED PEER-REVIEWED PUBLICATIONS

O'Rourke, H. P., & Hilley, C. D. (2025). A guide to specifying effects in latent change score models with moderated mediation. *Journal of Behavioral Data Science*, 25(2), 1–27.

O'Rourke, H. P., & Han, D. E. (2023). Considering the distributional form of zeroes when calculating mediation effects with zero-inflated count outcomes. *Journal of Behavioral Data Science*, 3(2), 1–14.

O'Rourke, H. P., Fine, K. L., Grimm, K. J., & MacKinnon, D. P. (2022). The importance of time metric precision when implementing bivariate latent change score models. *Multivariate Behavioral Research*, 57, 561–580.

Hilley, C. D., & **O'Rourke, H. P.** (2022). Dynamic change meets mechanisms of change: Mediation in the latent change score framework. *International Journal of Behavioral Development*, 46, 125–141.

Grimm, K. J., Helm, J., Rodgers, D., & **O'Rourke, H. P.** (2021). Analyzing cross-lag effects: A comparison of different cross-lag modeling approaches. *New Directions for Child and Adolescent Development*, 175, 11–33.

O'Rourke, H. P., & Vazquez, E. (2019). Mediation analysis with zero-inflated substance use outcomes: Challenges and recommendations. *Addictive Behaviors*, 94, 16–25.

SELECTED RESEARCH PRESENTATIONS

Brewer, G. A., & **O'Rourke, H. P.** (2026, July). *Exploring between-person and within-person variation in prospective memory*. Paper accepted at the 6th International Conference on Prospective Memory, Heidelberg, Germany.

O'Rourke, H. P. (2025, September). *From modeling linearity to machine learning: What's hot (and not) in the last century of data analysis in psychology*. Department of Psychology Annual Research Conference, UC Riverside.

O'Rourke, H. P. (2023, August). *Building superior multivariable systems to enhance precision in models of behavior change*. Presentation for Fralin Biomedical Research Institute at VTC, Virginia Tech, Roanoke, VA.

O'Rourke, H. P. (2020, September). *Mediation with zero-inflated count outcomes for substance use data*:

Negative binomial, Poisson, and hurdle models. McGill Quantitative Psychology Seminar, Montreal, Quebec, CA.

CURRENT EXTERNAL FUNDING

2023–2026: *Measuring Change in the Relations Amongst Cognitive, Health, and Socioemotional Factors at the Single-Subject Level and Improving Prediction of Critical Outcomes* (Army Research Institute / Army Materiel Command). Gene Brewer (PI), Holly O'Rourke (Co-PI). **\$1,707,828** (49% recognition)

2024–2027: *Active Learning at Scale: Transforming Teaching and Learning via Large-Scale Learning Science and Generative AI* (Institute of Education Sciences). Danielle McNamara (PI), Arner, Brewer, Craig, Roscoe, Holly O'Rourke (Consultant). **\$3,750,000** (5% recognition)

CURRENT INTERNAL FUNDING

2025–2026: *Improving Data Processing in Pursuit of Examining Intraindividual Variability in Cognition, Behavior, and Biological Health* (UCR CoR). O'Rourke (PI), Brewer (Co-I). **\$4,000**

SELECTED COMPLETED FUNDING

2019–2021: *Using Computational Linguistics to Detect Comprehension Processes in Constructed Responses across Multiple Large Data Sets (Resubmission)* (DoEd). Danielle McNamara (PI), Holly O'Rourke (Co-PI), Renu Balyan (Co-PI). **\$600,001** (10% recognition)

2019–2023 *iSTART-Early: Interactive Strategy Training for Active Reading and Thinking for Young Developing Readers (Resubmission)* (DoEd). Danielle McNamara (PI), Holly O'Rourke (Co-PI). **\$1,400,000** (10% recognition)

2020–2022: *Multi-Component Behavioral Intervention Designed to Increase Functional Independence during Aging in ASD (Resubmission)* (DoD). B. Blair Braden (PI), Holly P. O'Rourke (Co-I). **\$314,208** (30% recognition)

2018: *Interventions to Improve Divergent Thinking* (Laboratory for Analytic Sciences). Gene Brewer (PI), Holly O'Rourke (Co-PI), Adam Cohen (Co-I). **\$84,991** (40% recognition)

TEACHING EXPERIENCE

UC Riverside

PSYC 259: Statistical Mediation Analysis (graduate)

PSYC 109: Advanced Research Methods and Statistics (undergraduate)

Arizona State University

FAS 598: Quantitative Methods in the Social Sciences I, FAS 508: Structural Equation Modeling (graduate)

SOC 390: Social Statistics I, FAS 498: Advanced Statistics for Social Sciences (undergraduate)

EDITORIAL AND REVIEW SERVICE

Editorial Board, Statistical Advisor: *BJPsych Open*

Review Editor, 2020–2025: *Frontiers in Psychology: Quantitative Psychology and Measurement*

Ad Hoc Reviewer: *Multivariate Behavioral Research, Structural Equation Modeling, Behavior Research Methods, Computational Brain & Behavior, Frontiers in Mathematics and Statistics, Addictive Behaviors, Prevention Science, Psychology of Addictive Behaviors, Child Development, Practical Assessment, Review, and Evaluation*, and others

Ad Hoc Grant Reviewer: National Science Foundation (NSF)

TECHNICAL EXPERTISE

Languages: R, SAS, Python, MATLAB, C/C++

Software: R, SAS, Mplus, MATLAB, Mathematica, SPSS

Systems: Windows, Macintosh, Unix, HPC/cloud computing